Efficient Object Detection in Remote Sensing Images Based on Feature Weaving and Redundancy Suppression

Yunze Bai, Changming Song, Yun Wang, and Peiyan Li

***Abstract*—Remote sensing object detection faces significant challenges in handling scale variations, complex backgrounds, and dense target distributions. Existing models often struggle with high computational costs and a massive number of parameters, hindering deployment on resource-limited devices. To address these issues, we propose an efficient object detection network based on feature weaving and redundancy suppression called EWS-YOLO, featuring two components: 1) the weave-gated parallel (WGP) module employs grouped convolution with interlaced cascading and gating mechanisms to achieve implicit multiscale feature extraction and adaptive fusion, significantly enhancing multiscale target detection capability; and 2) the redundancy-suppressed feature pyramid network (RFS-FPN) integrates an progressive shuffle upsampling (PSU) mechanism coupled with dynamic feature fusion (DFF), enabling intelligent fusion of low-level spatial details and high-level semantic features while effectively suppressing redundant information through adaptive channelwise feature recalibration. Experimental results on VisDrone, DIOR, and UAVDT datasets demonstrate EWS-YOLO's superior performance, achieving 42.4%, 95.2%, and 98.7% mean average precision (mAP)@0.50, respectively, while reducing parameters by 41% compared to YOLOv8s, making it particularly suitable for resource-constrained applications.**

***Index Terms*—Dynamic feature fusion (DFF), feature weaving, object detection, progressive shuffle upsampling (PSU), redundancy suppression.**

I. Introduction

IN RECENT years, as a core component of remote sensing technology applications, remote sensing image object detection has played an indispensable role in military reconnaissance [1], environmental monitoring [2], urban planning [3], and other fields. However, due to the unique characteristics of remote sensing images, object detection in this domain faces multiple challenges. First, significant variations in object scales make it difficult for traditional methods to achieve balanced detection accuracy for targets of different sizes [4]. Second, complex background interference leads to weakened target features, significantly increasing the risks of false positives and missed detections [5].

In addition, densely distributed objects often result in mutual occlusion, where overlapping bounding boxes and redundant background information severely degrade detection accuracy [6], as shown in Fig. 1.

Current mainstream real-time remote sensing object detection models, such as the YOLO series [7], [8], [9], have made remarkable progress but still suffer from two key limitations. On the one hand, in excessive computational resource consumption, models such as YOLOv8s have a large number of parameters, requiring substantial computing power for training and inference, making them difficult to deploy efficiently on resource-constrained devices such as drones and satellites [10]. On the other hand, the feature fusion mechanisms are not sufficiently intelligent—existing methods often rely on simple concatenation and convolutional operations when handling multiscale features [11], failing to fully exploit the complementary relationships between different hierarchical features or effectively suppress feature redundancy [12]. These issues significantly hinder the application of remote sensing object detection in edge computing scenarios.

To address these challenges, we propose an efficient object detection network based on feature weaving and redundancy suppression (EWS-YOLO), which primarily consists of two core modules: the weave-gated parallel (WGP) module and the redundancy-suppressed feature pyramid network (RFS-FPN). The WGP module builds upon a single input feature map, utilizing convolutional operations to construct internally parallel multibranch structures, implicitly achieving multiscale feature extraction. Furthermore, by incorporating an interlaced cascading connection mechanism and a gating regulation mechanism, the module enables adaptive fusion of features at different hierarchical levels based on their importance, thereby significantly enhancing the model's ability to represent multiscale targets.

On the other hand, the RFS-FPN module adopts a progressive shuffle upsampling (PSU) strategy and a dynamic feature fusion (DFF) method to effectively integrate low-level spatial details with high-level semantic information. In addition, the module introduces an adaptive channelwise feature recalibration mechanism, which dynamically evaluates and weights the importance of each feature channel, significantly suppressing the interference of redundant features on detection performance.

The main contributions of this article are summarized as follows.

1) We propose the WGP module, which adopts a multibranch strategy to establish parallel processing, implicitly

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Fig. 1. Challenges of remote sensing object detection are influenced by various factors, including densely distributed, complex background, and large-scale differences.

extracting multiscale feature representations from a single input feature map. This design implements an interlaced cascading structure on partial channels, combined with gating units to dynamically adjust feature fusion weights, achieving adaptive fusion of multilevel features. This mechanism significantly enhances the model's ability to recognize multiscale targets while effectively reducing network parameters and improving the efficiency and flexibility of feature representation.

2) We construct an RFS-FPN that achieves efficient and synergistic fusion of low-level spatial details and high-level semantic features through the PSU mechanism and DFF technology. This design adopts a multibranch parallel processing architecture, utilizing feature addition instead of concatenation to reduce redundant convolutional computations, effectively suppressing feature redundancy while significantly decreasing the number of parameters.

3) We conducted systematic experiments on three authoritative remote sensing datasets, VisDrone [13], DIOR [14], and UAVDT [15], comparing the proposed EWS-YOLO with current mainstream detection methods. The experimental results demonstrate that EWS-YOLO achieves outstanding performance in detection accuracy. Meanwhile, EWS-YOLO reduces the number of parameters by 41% compared to YOLOv8s, fully validating the significant advantage of this algorithm in balancing accuracy and efficiency.

The remaining sections are organized as follows. Section II reviews related works, while Section III describes the details of the proposed model. In Section IV, we provide implementation details and experimental results. Finally, the conclusion is summarized in Section V.

II. Related Works

A. Object Detection in RSIs

In recent years, deep learning methods have made significant progress in the field of object detection. These methods rely on deep neural networks to extract features from remote sensing images and employ one- [16] or two-stage [17] strategies to predict target locations and categories. However, despite continuous performance improvements, existing detectors still face significant challenges in handling multiscale targets. In remote sensing images, detection accuracy is considerably limited due to factors such as significant target variations, complex background interference, and dense target distribution. To address this issue, Wang et al. [18] proposed Gold-YOLO, which achieves efficient cross-layer feature fusion through a gather-and-distribute mechanism and optimizes multiscale detection performance with a dual-branch structure. Dong et al. [19] introduced CSWinTransformer, leveraging a cross-shaped window self-attention mechanism to perform parallel computations on horizontal and vertical strips and adapting to inputs of varying sizes through locally enhanced positional encoding, thereby improving detection accuracy. Qian et al. [20] proposed a unified transferring strategy (UTS), which facilitates the transfer of bounding box regression (BBR) loss from horizontal object detection (HOD) to oriented object detection (OOD) by converting the BBR of oriented bounding box (OBB) into the joint BBR of its horizontal smallest enclosing rectangle (HSER) and two offsets. Qian et al. [12] proposed a weakly supervised object detection framework, which introduces a filtering and weighted synthesis guided by a category confidence score (FWSC) strategy to generate high-quality pseudo-ground truth (PGT) by removing low-confidence instances and synthesizing overlapping ones. Zhang et al. [22] proposed a cascaded autoregressive framework, which generates high-quality remote sensing images by treating objects as "tokens" and the background as a "prompt," where the model autoregressively predicts object attributes (e.g., positions, sizes, and orientations) and synthesizes object instances through a conditional reverse diffusion process. On the other hand, high-precision models often come with a large number of parameters, making them difficult to deploy on edge devices. To tackle this challenge, Ma et al. [23] proposed Starnet, which achieves implicit high-dimensional mapping and strong nonlinear expression through star operations, breaking

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the reliance of traditional lightweight networks on width expansion and activation functions. Han et al. [24] introduced GhostNet, which generates redundant ghost feature maps via linear transformations of intrinsic feature maps, significantly reducing computational overhead. However, existing methods often trade increased parameters for improved accuracy or sacrifice performance to reduce parameters. Therefore, the key to achieving high detection accuracy with low parameter counts lies in enhancing the efficiency of feature extraction and fusion.

B. Feature Pyramid Networks

The FPN enhances multiscale object detection performance by integrating shallow and deep features from different levels through a top-down pathway and lateral connections, constructing a pyramid structure rich in both semantic and spatial detail information. However, this architecture still faces challenges such as feature misalignment, information loss, limited spatial perception, and the difficulty of balancing computational efficiency with detection accuracy [25]. To address these issues, Tan et al. [26] proposed BiFPN, which enables bidirectional interaction between features of different resolutions through bidirectional cross-scale connections and introduces learnable weight parameters to dynamically adjust the contribution of different input features, thereby enhancing effective features while suppressing noise interference. Shi et al. [27] introduced HS-FPN, a model that leverages the strong semantic information of high-level features to dynamically filter spatial details in low-level features, further improving detection accuracy with high-frequency perception and spatial dependence modules. Nevertheless, existing FPN optimization methods often rely heavily on concatenation and convolution operations during feature fusion, leading to increased redundant features and reduced computational efficiency. Even approaches such as AFPN, proposed by Yang et al. [28], which employ progressive fusion strategies to enable direct cross-level information transfer and structural optimization, fail to fully resolve the issue of feature redundancy, resulting in diminished fusion efficiency. Therefore, a key factor in improving feature fusion efficiency lies in effectively suppressing redundant features.

In addition, most current models use upsampling methods such as bilinear interpolation [29] or nearest-neighbor interpolation [30], which, while convenient, are inefficient and struggle to meet the demands of multiscale feature fusion in terms of expressive power and computational performance. In summary, optimizing the feature fusion process requires a focus on suppressing redundant features and refining upsampling strategies to achieve a better balance between efficiency and accuracy.

III. Methodology

This section offers a comprehensive portrait of our proposed EWS-YOLO. We first present the overall architecture of the model, followed by WGP and RFS-FPN individually. Last, the loss functions utilized within the proposed model are summarized.

A. Overview

The overall architecture of the proposed model is depicted in Fig. 2. In general, the model comprises three parts: the backbone, RFS-FPN, and the detection head.

The EWS-YOLO model's backbone network consists of five feature extraction stages, from Stage_0 to Stage_4. The first three stages adopt the YOLOv8 module structure, composed of Conv2d-BatchNorm2d-Silu (CBS) and C2F (a residual module incorporating CBS, BottleNeck, and Concat). In Stage_3 and Stage_4, the original C2F modules are replaced with the proposed WGP module to achieve feature weaving. The RFS-FPN enables multiscale feature fusion through bidirectional paths and lateral connections, where the CBS module handles downsampling, the proposed PSU module optimizes upsampling via a progressive shuffling strategy, and the DFF module reduces parameters and suppresses redundancy through adaptive weighted fusion. Finally, the detection head employs a dual-branch structure to simultaneously accomplish target classification and localization.

B. Weave-Gated Parallel

In deep feature extraction networks (e.g., Stage_3 and Stage_4), the expansion of receptive fields often introduces substantial background noise, which, when integrated into feature maps, significantly degrades subsequent feature extraction and target detection performance [31]. To address this issue, we propose the WGP module illustrated in Fig. 3(a). The design of the WGP model draws inspiration from the multipathway processing mechanisms of the biological visual system, enhancing detection performance by constructing a multibranch architecture. Its core innovations lie in the introduction of the attentional hybrid computation (AHC) module and a dynamic drop mechanism. The AHC module employs head-shuffling technology to simultaneously achieve global context modeling and cross-branch information interaction without significantly increasing computational overhead, thereby enhancing the model's nonlinear expressive capability. Concurrently, the drop mechanism dynamically activates or deactivates the AHC branch based on the characteristics of the input sample, effectively suppressing feature redundancy and mitigating the risk of overfitting. These two mechanisms work in synergy with other pathways, collectively strengthening the model's representational efficiency and generalization ability.

The module divides input channels into three parallel branches (the number of channels is 0.5 multiples, 0.375 multiples, and 0.125 multiples of the input channels, respectively) to achieve hierarchical feature weaving. The first branch defined $B_{(i,1)}(i = 0, 1; c = 0.5)$ preserves raw information, establishing a lossless gradient propagation path to ensure model stability and performance baselines. The second branch defined $B_{(i,2)}(i = 0, 1; c = 0.375)$ employs a compression-expansion strategy, utilizing two Conv3 $\times$ 3 convolutions to enhance feature representation. The third branch defined $B_{(i,3)}(i = 0, 1; c = 0.125)$ incorporates an AHC module, which adaptively reweights feature maps via global context modeling, while a drop module is embedded to prevent overfitting. All branches adopt residual connections [32], with the second

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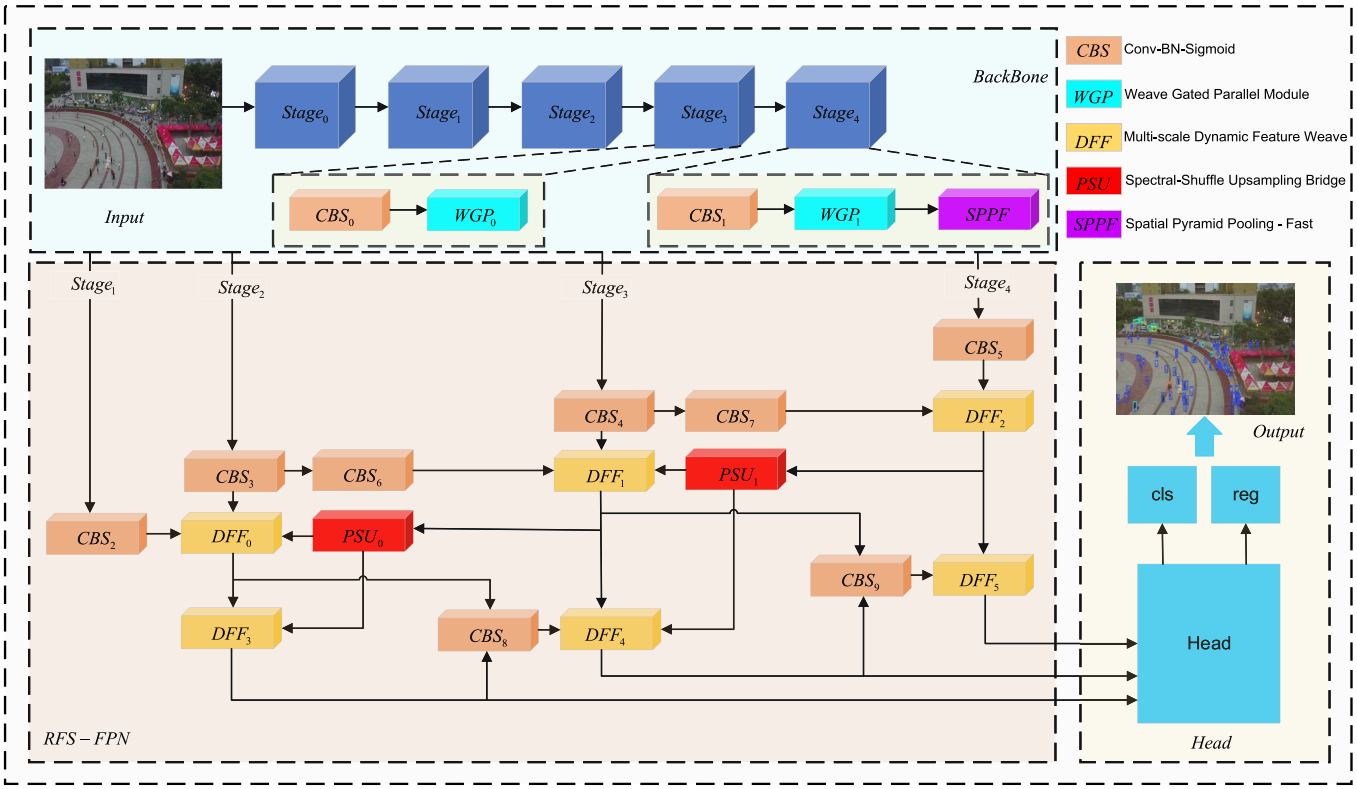


Fig. 2. Overall architecture of the proposed model.

and third branches connected via add operations to reduce parameter redundancy, collectively forming a residual topology centered on the first branch. This design ensures smooth gradient propagation and training stability

$$B_{(i,1)} = f_{\text{Conv}}^{1\times1}(\text{CBS}_{(i,c)}) \tag{1}$$

$$B_{(i,2)} = A(f_{\text{Conv}}^{3\times3}(f_{\text{Conv}}^{3\times3}(f_{\text{Conv}}^{1\times1}(\text{CBS}_{(i,c)}))), f_{\text{Conv}}^{1\times1}(\text{CBS}_{(i,c)})) \tag{2}$$

$$B_{(i,3)} = A(\text{Dp}(\text{AHC}(N(f_{\text{Conv}}^{1\times1}(\text{CBS}_{(i,c)})))), f_{\text{Conv}}^{1\times1}(\text{CBS}_{(i,c)})) \tag{3}$$

$$\text{WGP}_i = f_{\text{Conv}}^{1\times1}(\text{Concat}(\text{Concat}(B_{(i,2)}, B_{(i,3)}), B_{(i,1)})) \tag{4}$$

where N indicates nn.LayerNorm(Norm), A denotes the elementwise addition (add), Dp means drop, and c represents the multiple of the input channel.

AHC: The Transformer model [33], with its powerful long-range dependence modeling and global context awareness, demonstrates significant advantages in large object detection. However, it still faces challenges in small object detection, such as high computational complexity, attention dispersion caused by long sequences, and redundancy in attention patterns among different heads in multihead self-attention, which lack a specialized division of labor.

To address these challenges, we propose the AHC module, as illustrated in Fig. 3(b). First, the feature maps obtained from the convolutional operation are reshaped to generate query (Q), key (K), and value (V) tensors of the same shape, as shown in (5). Head_s denotes the number of attention heads used for Q, K, and V, which is set to 8 (denoted as Q_{Head_1}–Q_{Head_8}, K_{Head_1}–K_{Head_8}, and V_{Head_1}–V_{Head_8}, respectively); $\text{Channel}_{\text{Heads}}$ represents the number of channels corresponding to each attention head. After generating Q, K, and V, the module performs an explicit head shuffle operation to transpose and reorganize the outputs of different attention heads into Q_s, K_s, and V_s, where the operation involves swapping Head_s and $\text{Channel}_{\text{Heads}}$ in dimensions, as shown in (6). This forces interhead information interaction and fusion, breaking the homogeneity of attention patterns and significantly enhancing diversity. As a result, the model can better understand context from multiple perspectives and capture richer feature relationships, effectively improving the detection performance for small- and medium-sized objects. Furthermore, after the Attention computation, residual connections and depthwise separable convolutions are employed to enhance critical features while suppressing noise as EAttention. To finely regulate feature importance, a gated filtering mechanism is introduced subsequently, where the control signal from Linear adjusts the output of the residual layer, adaptively amplifying key features and suppressing noise and redundant information. Depthwise separable convolutions are also utilized to reduce parameter counts, thereby enhancing the model's perception of critical features and overall performance

$$\text{Shape}\,(Q, K, V) = (B, \text{Head}_s, \text{Channel}_{\text{Heads}}, H \times W) \tag{5}$$

$$\text{Shape}\,(Q_S, K_S, V_S) = (B, \text{Channel}_{\text{Heads}}, \text{Head}s, H \times W) \tag{6}$$

$$\text{Attention} = \text{Soft max}\left(\frac{Q_s K_i^T}{\sqrt{d_k}}\right) V_S \tag{7}$$

$$\text{EAttention} = f_{\text{Conv}}^{1\times1}\left(A\left(\text{Attention}, f_{\text{Conv}}^{1\times1}\left(\text{Attention}\right)\right)\right) \tag{8}$$

$$\text{AHC} = D\left(\text{GELU}\left(f_{\text{Conv}}^{3\times3}\left(\text{EAttention}\right)\right) \times \text{Linear}\left(\text{EAttention}\right)\right) \tag{9}$$

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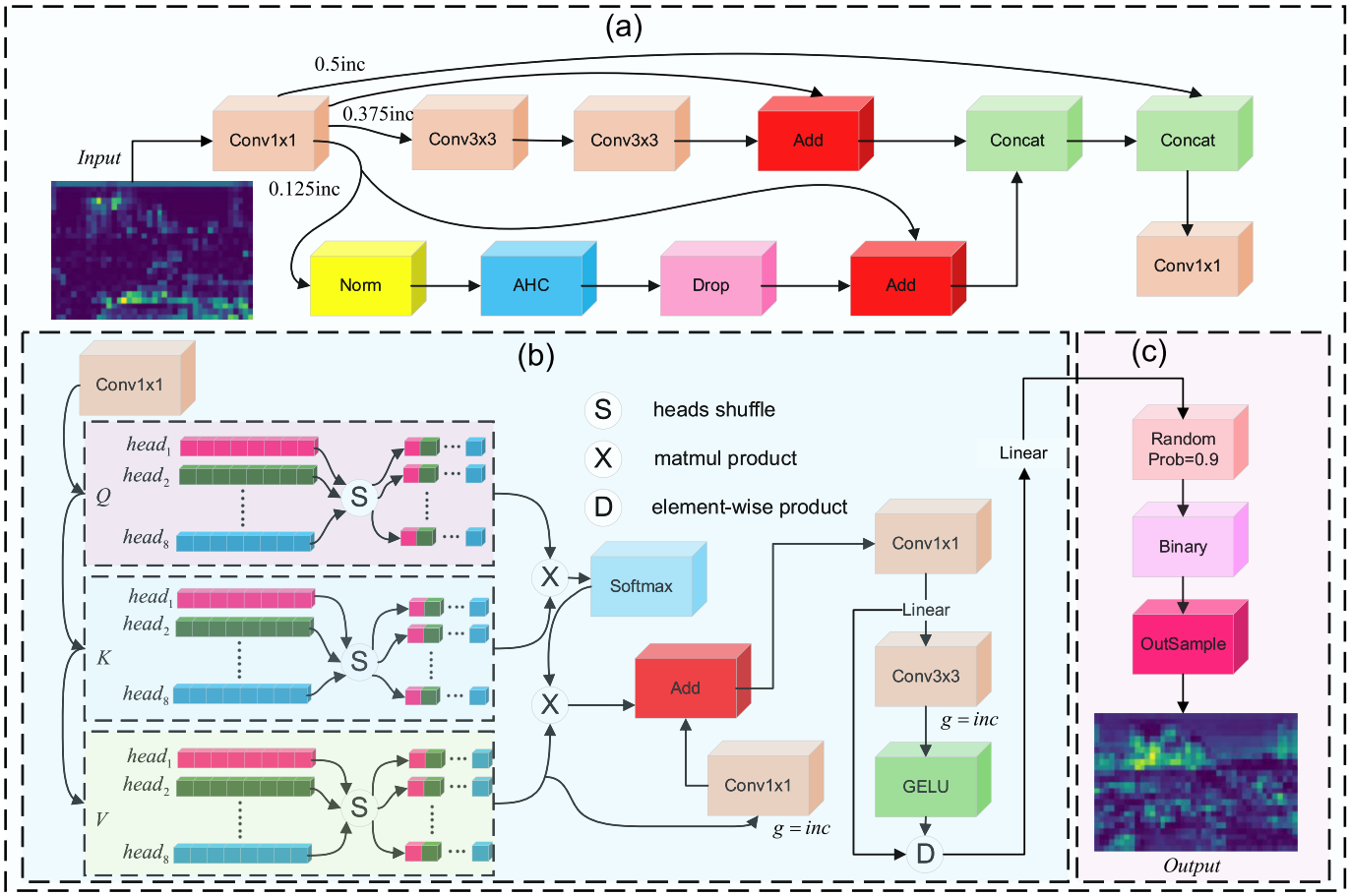


Fig. 3. Structure of the AHC. (a) WGP. (b) AHC. (c) Drop.

where d_k means the scaling factor, A and D indicates the elementwise addition and the elementwise product, respectively, GELU [34] denotes the activation function, and Linear represents nn.linear.

Drop: As shown in Fig. 3(c), our drop module sets the retention probability to 0.9, first generates a random mask in the range [1], and then determines whether to discard the current path through rounding down and binarization operations. Unlike traditional Dropout [35] that randomly masks neurons, this method directly chooses random dropping of samples, effectively reducing the number of samples involved in computation. This mechanism can be regarded as an implicit curriculum learning strategy, gradually transitioning the training process from complete samples to partial samples, dynamically adjusting the learning difficulty while significantly reducing redundant computation. Furthermore, by preventing the model from overrelying on any specific data point, it further enhances the suppression of overfitting.

C. Redundancy-Suppressed FPN (RFS-FPN)

The FPN is a multiscale feature fusion method for object detection and semantic segmentation [36]. Its core idea lies in integrating low-level high-resolution features with high-level strong semantic information through a bottom-up, top-down, and lateral connection structure, thereby effectively improving the detection performance for multiscale objects (especially small targets) while maintaining high computational efficiency. However, FPN typically performs feature fusion only at Stage_2, Stage_3, and Stage_4, which still leaves room for improvement in the detection accuracy of small targets. Although incorporating Stage_1 can supplement finer details, it significantly increases computational overhead. In addition, channel concatenation during fusion may introduce aliasing effects [37], leading to information redundancy and reduced localization accuracy. To address these issues, we propose the RFS-FPN architecture, as illustrated in Fig. 1. By downsampling Stage_1 to provide richer detailed information and designing an efficient upsampling module (PSU) that progressively performs upsampling and incorporates channel shuffling to enhance feature representation, we achieve improved performance. Furthermore, we introduce a DFF module, which combines dynamic self-attention mechanisms, multiscale depthwise separable convolutions, and feature shuffling [38]. This module significantly reduces parameters and redundant information while markedly boosting detection accuracy. Finally, to ensure smooth information flow, we integrate residual connections in all DFF components and the PSU module, preserving low-level detail features and, thereby, enhancing network depth and expressive power. This optimization improves overall system performance and training stability.

PSU: The proposed PSU module is illustrated in Fig. 4. Its design is based on low-computation deep separable

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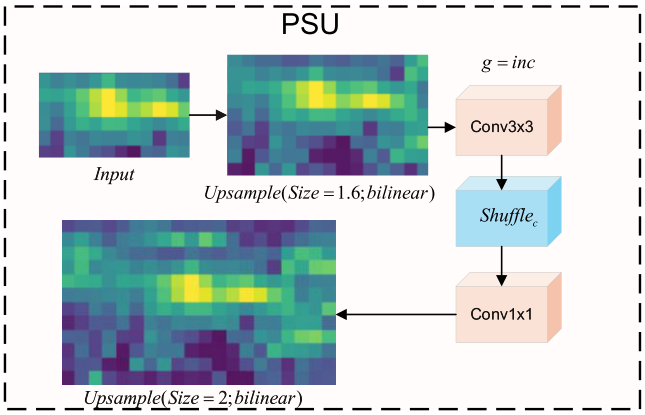


Fig. 4. Structure of the PSU.

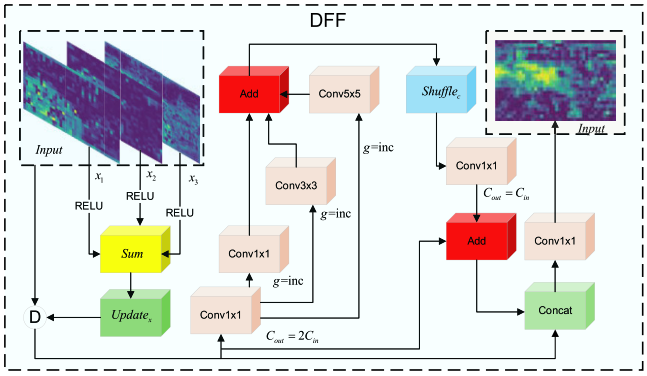


Fig. 5. Structure of the DFF.

convolutions, combined with dynamic shuffling technology. The core concept is to break away from the limitations of traditional fixed upsampling patterns. It enhances output quality by dynamically boosting feature expression capabilities and achieves a balance through a lightweight design that also considers computational efficiency. PSU introduces a learnable 3×3 depthwise separable convolutional layer [39] and a 1×1 convolution between two upsampling operations, which adaptively extracts and processes features while significantly reducing the computational overhead compared to standard convolution. The shuffle module enhances interchannel information interaction and fusion by rearranging feature map elements across different channels, breaking channel independence, and enabling more efficient utilization of spatial information. This improves the spatial consistency and quality of upsampled feature maps, facilitating more accurate localization and recognition in subsequent networks. In addition, the progressive upsampling approach mitigates geometric distortion and errors caused by single-step large-scale magnification, allowing intermediate feature correction and optimization to produce smoother and more precise high-resolution outputs

$$U_{(i,1)} = \text{Upsample}_{1.6}\left(\text{DFF}_i\right)(i = 1, 2) \tag{10}$$

$$S_i = f_{\text{Conv}}^{1\times1}\left(\left(\text{Shuffle}\left(f_{\text{Conv}}^{3\times3}\left(U_{(i,1)}\right)\right)\right)\right) \tag{11}$$

$$U_{(i,2)} = \text{Upsample}_2\left(S_i\right) \tag{12}$$

$$\text{PSU}_i = U_{(i,1)} \rightarrow S_i \rightarrow U_{(i,2)} \tag{13}$$

where $U_{(i,1)}, S_i$, and $U_{(i,2)}$, respectively, represent the first upsampling, the shuffle operation, and the second upsampling. $\text{Upsample}_{1.6}$ represents 1.6 times upsampling.

DFF: The proposed DFF module is illustrated in Fig. 5. The core design of this model adheres to the principles of efficient fusion and strong representation. It employs a DFF mechanism to adaptively integrate features from different levels in a learnable manner, significantly reducing information redundancy. Simultaneously, multiscale depthwise separable convolutions are introduced to extract multiscale features in parallel, enhancing the model's representational capacity for targets of varying sizes. Furthermore, a feature shuffling strategy is incorporated to promote information interaction across channels.

For the input multiscale feature maps $X_i (i = 1 \text{ to } n)$, we introduce a trainable parameter x_i for each feature map to dynamically learn their importance weights. These weights are then normalized and updated by the Update_x module to obtain x_i' and multiplied elementwise with the original feature maps through a residual structure to generate multiscale adaptive feature maps M. To further enhance feature representation, a parallel multibranch architecture is employed: The first branch P_1 sequentially applies 1×1, 3×3, and 5×5 depthwise separable convolutions of different scales, significantly improving the model's adaptability to complex scenes while substantially reducing parameters and computational costs. Channel shuffling is then performed to enhance cross-layer feature interaction, promoting multiscale information fusion while effectively avoiding redundancy and improving feature diversity and discriminability. The second branch P_2 preserves the original feature information to ensure that critical spatial details are retained. The entire module employs dual-path parallel processing with final fusion, featuring a compact and efficient structure. By extensively utilizing 1×1 convolutions and simple addition operations, it maintains powerful functional characteristics while ensuring computational efficiency, enabling flexible integration into various large-scale networks

$$x_i' = \text{Update}_x\left(x_i\right) = \frac{\text{RELU}\left(x_i\right)}{\sum \text{RELU}\left(x_i\right)} \tag{14}$$

$$M = P_2 = f_{\text{Conv}}^{1\times1}\left(D\left(x_i', X_i\right)\right) \tag{15}$$

$$P_1 = f_{\text{Conv}}^{1\times1}\left(\text{Shuffle}\left(A\left(f_{\text{Conv}}^{1\times1}\left(M\right), f_{\text{Conv}}^{3\times3}\left(M\right), f_{\text{Conv}}^{5\times5}\left(M\right)\right)\right)\right) \tag{16}$$

$$\text{DFF} = f_{\text{Conv}}^{1\times1}\left(\text{Concat}\left(A\left(P_1, P_2\right)\right)\right). \tag{17}$$

D. Loss Functions

We use the loss function of YOLOv8, which can be summarized as classification Loss [40] and BBR loss [41]

$$\text{Loss} = L + \text{DFL} + \text{CIoU} \tag{18}$$

where the classification loss is defined as follows:

$$L = \frac{1}{N}\sum_i L_i = \frac{1}{N}\sum_i -\left[y_i^* \log\left(p_i\right) + \left(1 - y_i\right)^* \log\left(1 - p_i\right)\right] \tag{19}$$

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where L represents the binary cross-entropy, N denotes the number of classes, y_i indicates ground-truth labels, and p stands for predicted probabilities.

The BBR loss is defined as DFL + CIoU [42], where the definition of distribution focal loss (DFL) [43] is given as follows:

$$\mathrm{DFL}\left(S_i, S_{i+1}\right) = -\left(\left(y_{i+1} - y\right)\log\left(S_i\right) + \left(y - y_i\right)\log\left(S_{i+1}\right)\right) \tag{20}$$

where S_i and S_{i+1} represent the predicted value and the adjacent predicted value output by the network, respectively, while y, y_i, and y_{i+1} denote the ground-truth value, the label integral value, and the adjacent label integral value of the target.

The definition of CIoU is given as follows:

$$v = \frac{4^*\left(\arctan\frac{w_G}{h_G} - \arctan\frac{w_P}{h_P}\right)^2}{\Pi^2} \tag{21}$$

$$\alpha = \frac{v}{(1 - \mathrm{IoU}) + v} \tag{22}$$

$$\mathrm{CIoU} = 1 - \mathrm{IoU} + \frac{d^2}{c^2} + \alpha v \tag{23}$$

where IoU represents the Intersection over Union, (w_G, h_G) and (w_P, h_P) denote the width and height of the predicted box and ground-truth box, respectively, and d represents the Euclidean distance between the center points of the predicted bounding box and the ground-truth bounding box, while c represents the diagonal length of the smallest enclosing box that can contain both the predicted and the ground-truth boxes, and α is a dynamic weighting coefficient that balances the contribution ratio between center point distance loss and aspect ratio loss.

IV. Experiment

A. Datasets

To comprehensively evaluate model performance, we conducted systematic experimental validation on two standard remote sensing datasets: VisDrone and DIOR.

1) *VisDrone Dataset:* The VisDrone dataset by Tianjin University's AISKYEYE team includes 10 249 UAV images and 288 videos, covering urban and rural scenes under varying conditions. It provides fine-grained annotations for ten object categories, with 30% targets under 32×32 pixels and dense occlusions. Supporting detection, tracking, and crowd counting, it is a UAV vision benchmark.

2) *DIOR Dataset:* The DIOR dataset, developed by Wuhan University, is a large-scale benchmark for remote sensing object detection. It addresses challenges such as scale diversity and complex backgrounds, covering 20 categories, including buildings and aircraft. It provides a standardized evaluation for algorithmic innovation in RS scenarios.

3) *UAVDT Dataset:* UAVDT is a large-scale benchmark dataset specifically designed for object detection from a UAV perspective. It contains approximately 80 000 frames of images covering various complex scenarios such as urban squares and main roads. The dataset primarily focuses on vehicle targets, making it suitable for fundamental computer vision tasks such as object detection and single object tracking.

TABLE I
Details of mAP Indicators

metric	detail
mAP	mAP at IoU =0.50 : 0.05 : 0.95
mAP@.50	mAP at IoU =0.50
mAP@.75	mAP at IoU =0.75
mAP@S	mAP for small objects: area $< 32^2$
mAP@M	mAP for medium objects: 32^2 <area< 96^2
mAP@L	mAP for large objects: area $> 96^2$

TABLE II
Details of mAR Indicators

metric	detail
mAR	mAR for all objects
mAR@S	mAR for small objects: area $< 32^2$
mAR@M	mAR for medium objects: 32^2 <area< 96^2
mAR@L	mAR for large objects: area $> 96^2$

B. Evaluation Metrics

In order to concisely assess and contrast the results of various detectors, four key metrics are employed: mean average precision (mAP), mean average recall (mAR), parameters, and giga floating-point operations per second (GFLOPs).

1) *mAP:* The mAP is a metric used to evaluate the precision of a detector, which can be subdivided into six types, as shown in Table I.

2) *mAR:* The mAR is a metric used to measure detector recall. To measure recall for all objects, including small, medium, and large objects, we utilize four refinement metrics: mAR, mAR@S, mAR@M, and mAR@L, which are presented in Table II.

3) *GFLOPs, Parameters, and FPS:* GFLOPs measure the computational complexity of a model, while the number of parameters reflects its storage requirements and expressive capacity. FPS stands for inference speed.

C. Implementation Details

Experiments were conducted on an Intel i9-9900X/NVIDIA GTX 4090 workstation using YOLOv8s as the baseline model. Training used SGD optimizer with cosine annealing learning scheduling strategy (lr = 0.01 and weight decay = 0.0005), mosaic data augmentation technology, 640×640 input resolution, batch size = 8, and 200 epochs for convergence. All models were trained from scratch using random initialization, without employing any pre-trained weights, to ensure fairness in model evaluation and comparability of the results.

D. Comparison Results

We compare our proposed EWS-YOLO against nine state-of-the-art object detection models on the VisDrone2019, DIOR, and UAVDT datasets: YOLOv8s, YOLOv10s, ASF-YOLO [44], Hyper-YOLO [45], GHOST-YOLO, GOLD-YOLO, LSKNet [46], Unireplknet [47], and Starnet. To meet the deployment requirements on resource-constrained

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TABLE III
QUANTITATIVE COMPARISON OF THE VISDRONE DATASET

Models	mAP	mAP@.50	mAP@.75	mAP@.S	mAP@.M	mAP@.L	mAR	mAR@S	mAR@M	mAR@L	param	GFLOPs	FPS
YOLOv8s	0.229	0.408	0.228	0.135	0.347	0.462	0.364	0.262	0.516	0.598	11.2M	28.6	328
YOLOv10s	0.230	0.414	0.229	0.140	0.342	0.444	0.375	**0.279**	0.517	0.656	8.1M	25.1	339
ASF-YOLO	0.220	0.398	0.221	0.128	0.337	0.412	0.353	0.258	0.499	0.544	8.9M	25.1	308
Hyper-YOLO	0.227	0.405	0.229	0.133	0.345	0.435	0.364	0.266	0.512	0.585	14.8M	38.9	189
GHOST-YOLO	0.195	0.359	0.191	0.109	0.297	0.396	0.332	0.231	0.481	0.524	6.0M	**16.4**	327
GOLD-YOLO	0.224	0.404	0.227	0.131	0.346	0.421	0.361	0.259	0.515	0.579	13.6M	29.9	187
LSKNet	0.166	0.316	0.161	0.081	0.253	0.365	0.300	0.199	0.439	0.489	10.3M	30.7	146
Unireplknet	0.208	0.379	0.205	0.116	0.318	0.404	0.350	0.247	0.500	0.530	10.5M	26.6	120
Starnet	0.181	0.341	0.177	0.098	0.279	0.381	0.316	0.214	0.464	0.491	6.6M	17.6	**342**
EWS-YOLO	**0.241**	**0.424**	**0.244**	**0.142**	**0.362**	**0.485**	**0.378**	0.276	**0.530**	**0.631**	**6.5M**	22.8	332
NanoDet	0.141	0.240	0.136	0.062	0.210	0.302	0.253	0.177	0.401	0.422	2.5M	5.9	19
PP-YOLO-TINY	0.149	0.263	0.140	0.068	0.232	0.311	0.274	0.180	0.429	0.483	4.2M	8.7	24
YOLOv8n	0.163	0.290	0.158	0.082	0.253	0.360	0.299	0.194	0.443	0.532	3.1M	8.9	25
YOLOv10n	0.159	0.285	0.155	0.083	0.246	0.315	0.296	0.198	0.436	0.458	2.8M	8.8	31
EWS-YOLOn	**0.191**	**0.328**	**0.190**	**0.102**	**0.293**	**0.430**	**0.327**	**0.226**	**0.473**	**0.577**	**1.9M**	**5.8**	**33**

TABLE IV
QUANTITATIVE COMPARISON OF THE DIOR DATASET

Models	mAP	mAP@.50	mAP@.75	mAP@.S	mAP@.M	mAP@.L	mAR	mAR@S	mAR@M	mAR@L	param	GFLOPs	FPS
YOLOv8s	0.674	0.944	0.748	0.354	0.681	0.879	0.718	0.489	**0.748**	0.912	11.2M	28.6	328
YOLOv10s	0.678	0.948	**0.757**	**0.379**	0.689	0.873	**0.723**	0.479	0.746	0.915	8.1M	25.1	339
ASF-YOLO	0.674	0.947	0.747	0.354	0.676	0.878	0.715	0.467	0.738	0.912	8.9M	25.1	308
Hyper-YOLO	0.663	0.937	0.739	0.361	0.669	0.865	0.707	0.462	0.734	0.901	14.8M	38.9	189
GHOST-YOLO	0.665	0.938	0.738	0.363	0.659	0.875	0.708	0.476	0.731	0.908	**6.0M**	**16.4**	327
GOLD-YOLO	0.676	0.949	0.757	0.360	0.679	0.874	0.715	0.483	0.735	0.908	13.6M	29.9	187
LSKNet	0.629	0.916	0.697	0.301	0.643	0.836	0.685	0.454	0.713	0.880	10.3M	30.7	146
Unireplknet	0.647	0.928	0.728	0.343	0.656	0.849	0.702	0.477	0.735	0.892	10.5M	26.6	120
Starnet	0.651	0.938	0.724	0.319	0.666	0.858	0.699	0.446	0.737	0.897	6.6M	17.6	**342**
EWS-YOLO	**0.684**	**0.952**	**0.757**	0.370	**0.694**	**0.889**	**0.723**	**0.496**	0.743	**0.918**	6.5M	22.8	332
NanoDet	0.553	0.832	0.616	0.238	0.549	0.731	0.661	0.419	0.669	0.811	2.5M	5.9	19
PP-YOLO-TINY	0.592	0.895	0.633	0.242	0.602	0.752	0.678	0.428	0.681	0.846	4.2M	8.7	24
YOLOv8n	0.632	0.911	0.705	0.317	0.627	0.836	0.689	0.432	0.729	0.878	3.1M	8.9	25
YOLOv10n	0.610	0.906	0.680	0.288	0.595	0.796	0.682	0.446	0.707	0.865	2.8M	8.8	31
EWS-YOLOn	**0.667**	**0.938**	**0.743**	**0.345**	**0.678**	**0.868**	**0.709**	**0.465**	**0.739**	**0.904**	**1.9M**	**5.8**	**33**

devices, we designed a lightweight version of EWS-YOLO, named EWS-YOLOn. Its core optimization involves reducing the network width by half, meaning that the number of channels in all convolutional layers is halved. To validate its effectiveness, we selected four current representative lightweight object detectors as benchmarks for comparison: NanoDet [48], PP-YOLO-Tiny, YOLOv8n, and YOLOv10n.

To accurately reflect performance comparisons across different hardware platforms, FPS testing for EWS-YOLO and its benchmark models was conducted on an NVIDIA GTX 4090 workstation, while the evaluation for the lightweight model EWS-YOLOn, designed for edge devices, and its corresponding comparison models was performed on a Jetson Nano embedded platform.

1) Results on the VisDrone Dataset: Table III presents the quantitative comparison results on the VisDrone dataset, where our EWS-YOLO model and EWS-YOLOn demonstrate significant improvements across most evaluation metrics. Compared to high-performance models (e.g., YOLOv10s), EWS-YOLO achieves higher overall detection accuracy in terms of mAP and mAR while maintaining fewer parameters and lower computational complexity. In contrast to lightweight models (e.g., YOLOv10n), EWS-YOLOn delivers the best detection accuracy with extremely low parameter counts, GFLOPs, and FPS, fully showcasing its efficient and balanced performance, and it has been proven that our lightweight version model can be fully deployed on edge devices.

Fig. 6 presents a visual comparison of detection results between baseline models and EWS-YOLO on the VisDrone dataset. EWS-YOLO demonstrates significant advantages in multiscale object recognition, with a notably higher number of successfully detected targets compared to other models. Particularly noteworthy is its stable performance in complex scenarios with severe occlusions, where it consistently identifies the majority of occluded objects, fully showcasing the generalization capability and robustness advantages of its feature-weaving architecture. However, our model still exhibits certain limitations. As indicated by the orange bounding boxes in the figure, partial small-scale targets remain undetected, highlighting the model's ongoing challenges in effectively handling small object detection tasks.

2) Results on the DIOR Dataset: Table IV presents the quantitative comparison results on the DIOR dataset, where the EWS-YOLO and EWS-YOLOn models demonstrate

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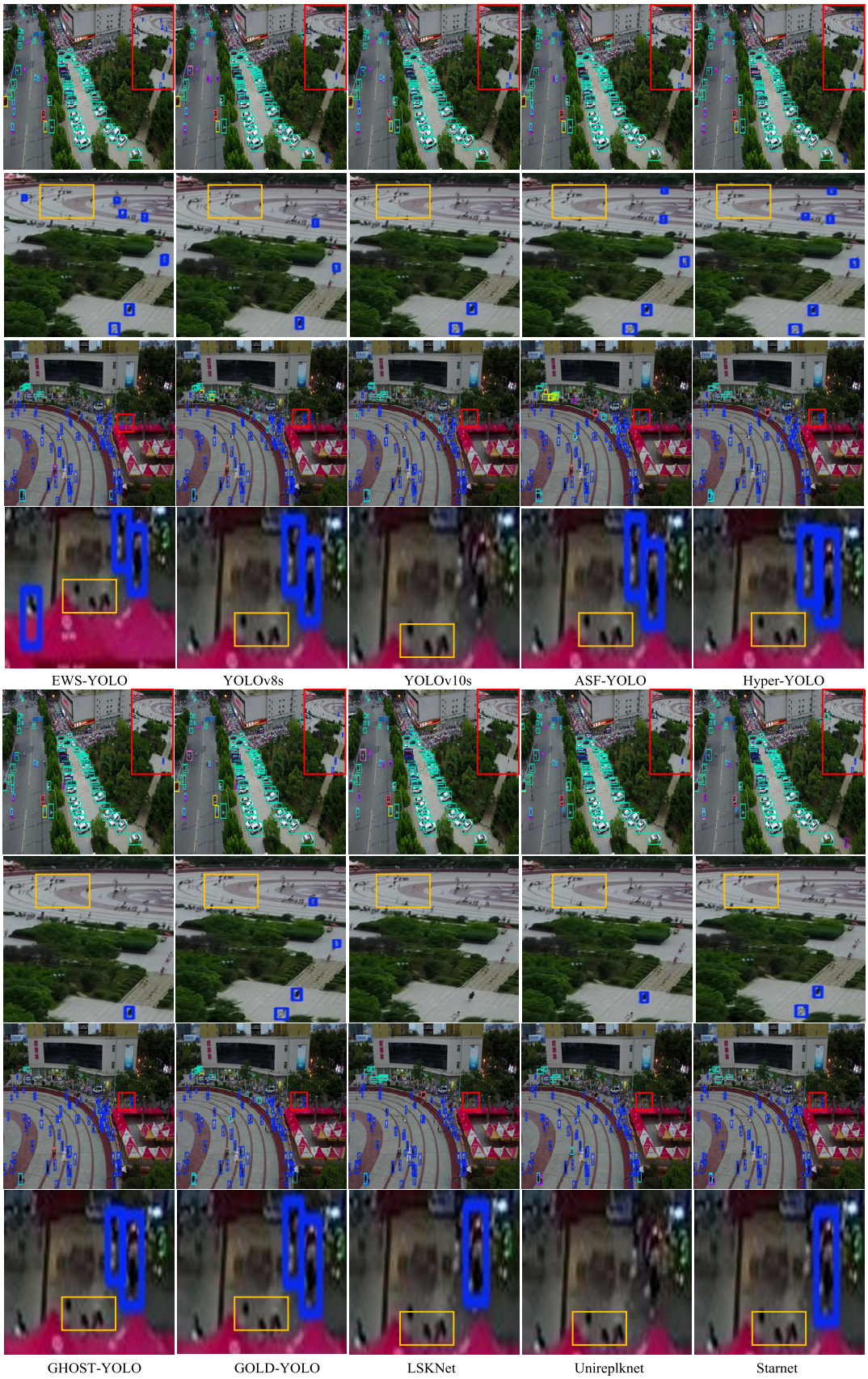


Fig. 6. Visualization of the predictions on the VisDrone dataset; the orange box indicates missed detection.

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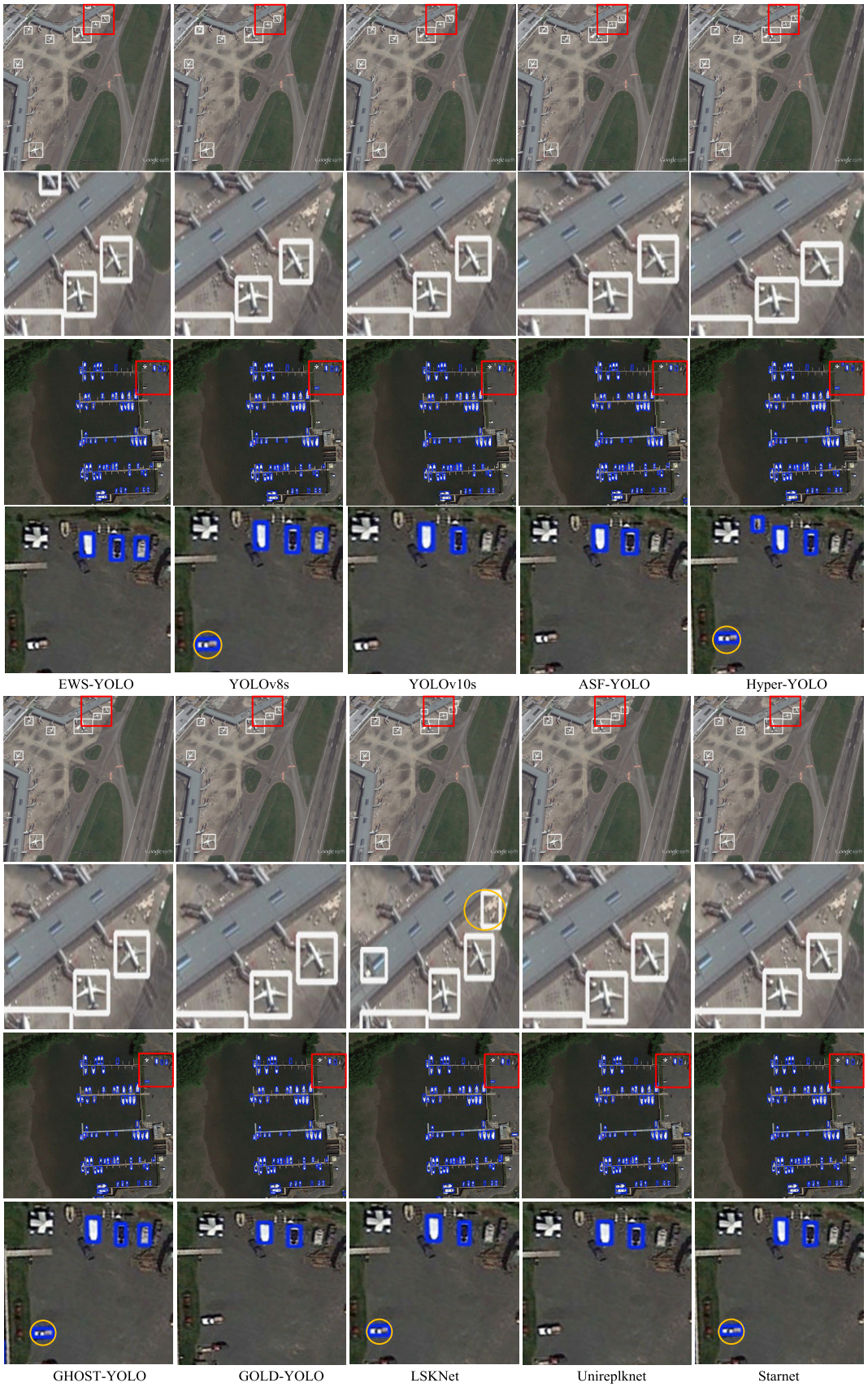


Fig. 7. Visualization of the predictions on the DIOR dataset, with the false detection indicated by the yellow circle.

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TABLE V
QUANTITATIVE COMPARISON OF THE UAVDT DATASET

Models	mAP	mAP@.50	mAP@.75	mAP@.S	mAP@.M	mAP@.L	mAR	mAR@S	mAR@M	mAR@L	param	GFLOPs	FPS
YOLOv8s	0.719	0.977	0.853	0.573	0.793	0.884	0.763	0.646	0.833	0.911	11.2M	28.6	328
YOLOv10s	0.682	0.968	0.813	0.545	0.756	0.791	0.743	0.637	0.807	0.862	8.1M	25.1	339
ASF-YOLO	0.720	0.974	0.853	0.568	0.801	0.851	0.765	0.641	0.840	0.891	8.9M	25.1	308
Hyper-YOLO	0.693	0.969	0.820	0.532	0.775	0.845	0.743	0.613	0.821	0.882	14.8M	38.9	189
GHOST-YOLO	0.693	0.970	0.823	0.534	0.776	0.863	0.744	0.616	0.818	0.891	**6.0M**	**16.4**	327
GOLD-YOLO	0.699	0.972	0.835	0.553	0.766	0.833	0.749	0.628	0.821	0.885	13.6M	29.9	187
LSKNet	0.617	0.924	0.724	0.427	0.713	0.813	0.685	0.539	0.769	0.856	10.3M	30.7	146
Unireplknet	0.636	0.945	0.744	0.457	0.728	0.789	0.707	0.577	0.782	0.866	10.5M	26.6	120
Starnet	0.660	0.953	0.774	0.487	0.746	0.832	0.718	0.585	0.795	0.872	6.6M	17.6	**342**
EWS-YOLO	**0.794**	**0.987**	**0.919**	**0.660**	**0.869**	**0.942**	**0.824**	**0.710**	**0.891**	**0.960**	6.5M	22.8	332
NanoDet	0.623	0.923	0.801	0.511	0.733	0.782	0.706	0.589	0.791	0.832	2.5M	5.9	19
PP-YOLO-TINY	0.659	0.926	0.810	0.530	0.748	0.793	0.727	0.614	0.799	0.870	4.2M	8.7	24
YOLOv8n	0.703	0.956	0.843	0.558	0.792	0.887	0.758	0.630	0.832	0.907	3.1M	8.9	25
YOLOv10n	0.688	0.945	0.819	0.547	0.760	0.792	0.748	0.635	0.811	0.887	2.8M	8.8	31
EWS-YOLOn	**0.747**	**0.973**	**0.881**	**0.602**	**0,824**	**0.904**	**0.788**	**0.664**	**0.861**	**0.929**	**1.9M**	**5.8**	**33**

significant advantages across most evaluation metrics. They achieve 68.4% and 66.7% in mAP and 72.3% and 70.9% in mAR, validating their outstanding performance in object detection tasks. Meanwhile, through feature redundancy suppression techniques, the models' parameters are compressed to only 6.5M and 1.9M, achieving higher detection accuracy while maintaining lightweight characteristics, fully meeting the dual requirements of high efficiency and low resource consumption for edge device deployment.

Fig. 7 presents a visual comparison of detection performance between various models and EWS-YOLO on the DIOR dataset. Experimental results demonstrate that EWS-YOLO exhibits significant advantages in handling occluded targets, outperforming other comparative models in both localization and classification accuracy. The model not only accurately identifies the contours and categories of occluded objects but also reduces both false detection and missed detection rates, fully validating its robust detection capability and efficient performance in complex scenarios.

Overfitting Analysis: The EWS-YOLO model demonstrates balanced and credible overall performance on the DIOR dataset. It achieves an mAP score of 0.684, which shows a noticeable decrease compared to its mAP@0.50 of 0.952—this decline aligns with the expected performance drop under stricter localization requirements. Meanwhile, at a higher IoU threshold of 0.75, its mAP remains competitive at 0.757, comparable to other high-performing models. The model exhibits stable precision and recall across objects of various sizes, with no significant signs of overfitting. Structurally, with only 6.5M parameters and 22.8 GFLOPs computational cost, it maintains a medium-low complexity, which helps avoid the risk of overparameterized models merely "memorizing" training data and further reinforces the reliability of its performance.

3) Results on the UAVDT Dataset: As shown in Table V, the EWS-YOLO model achieves the best overall performance on the UAVDT dataset. It significantly outperforms other comparative models across key detection metrics, including mAP (0.794), mAP@0.50 (0.987), and mAP@0.75 (0.919), while also demonstrating a good balance between computational efficiency and parameter count (parameters: 6.5M, GFLOPs: 22.8, and FPS: 332). The EWS-YOLO model shows particularly outstanding performance in small object detection (mAP@S: 0.660) and recall rate (mAR: 0.824), validating its robustness in complex aerial scenarios. The EWS-YOLOn, designed for resource-constrained scenarios, further compresses the model size, reducing parameters to 1.9M and computational cost to 5.8 GFLOPs. While maintaining relatively high detection accuracy (mAP: 0.747), it achieves a real-time inference speed of 33 FPS on edge devices such as the Jetson Nano, striking a good balance between lightweight design and performance. Although its accuracy is slightly lower compared to the original model, it still significantly surpasses similar lightweight models (e.g., YOLOv8n, YOLOv10n, and NanoDet) despite a substantial reduction in parameters, demonstrating the effectiveness of this lightweight design.

Fig. 8 presents a visual comparison of detection performance between various models and EWS-YOLO on the UAVDT dataset. The results demonstrate that EWS-YOLO achieves the highest detection accuracy under both daylight and low-light conditions. It is particularly notable for its superior ability to detect more occluded objects compared to other models, showcasing a significant advantage. In addition, EWS-YOLO maintains the lowest false detection rate among all models evaluated.

Overfitting Analysis: EWS-YOLO demonstrates outstanding performance on the UAVDT dataset, achieving a high mAP@0.50 of 0.973. This result can be attributed to the dataset containing only three target categories, which has a relatively simple structure and reduces the classification difficulty. Furthermore, the model's performance on metrics such as mAP and mAP@0.75 is close to its mAP@0.50, and it shows balanced results in mAP and mAR across different target scales, indicating no significant weaknesses. In addition, other mainstream models have also achieved similar mAP levels on this dataset. In conclusion, the risk of overfitting for EWS-YOLO can be essentially ruled out.

E. Ablation Experiments

1) Effect of WGP and RFS-FPN to EWS-YOLO: To validate the effectiveness of the WGP and RFS-FPN modules, we

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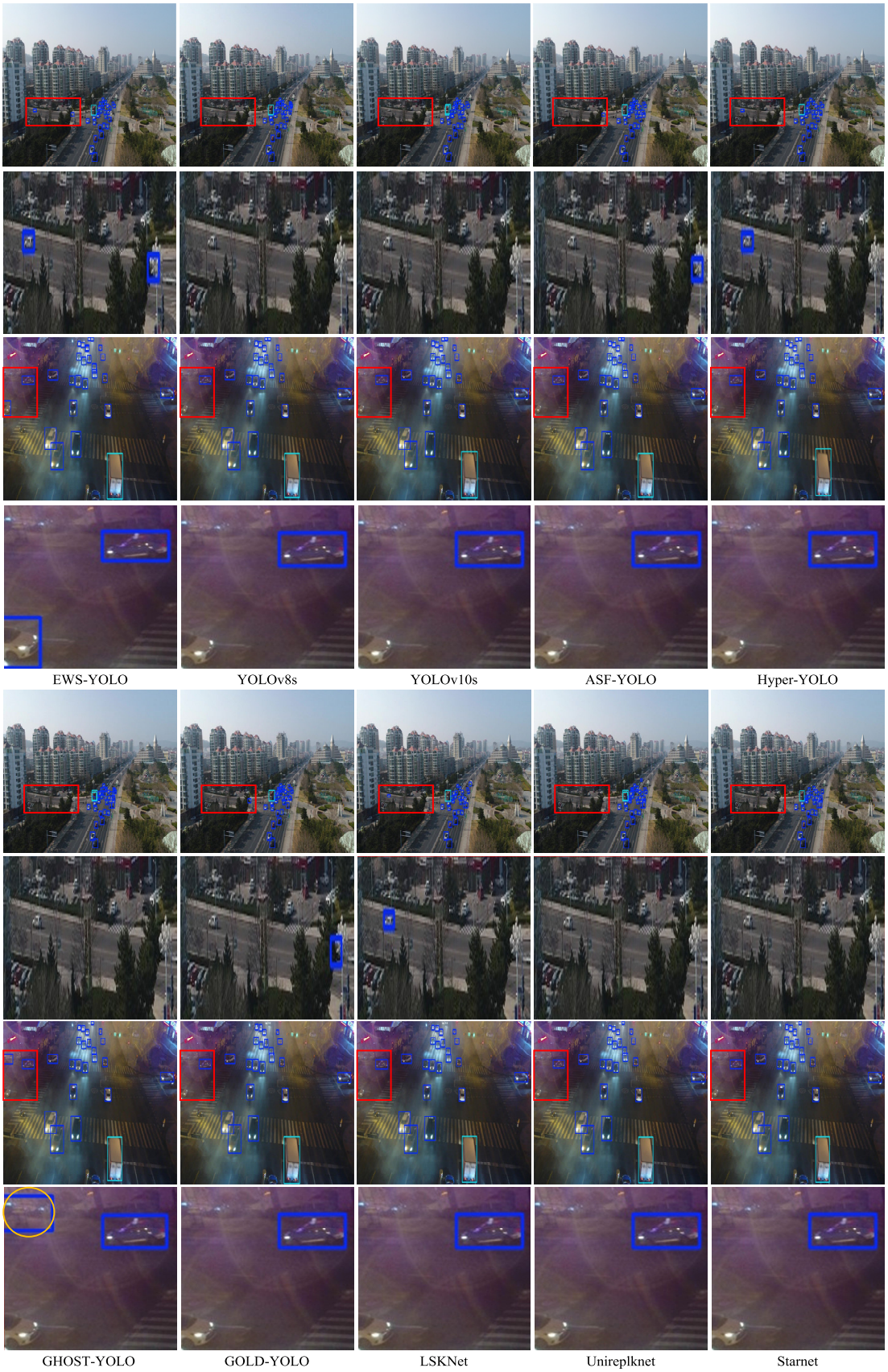


Fig. 8. Visualization of the predictions on the UAVDT dataset, with the false detection indicated by the yellow circle.

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TABLE VI
EFFECT OF WGP AND RFS-FPN ON THE VISDRONE DATASET

WGP	RFS-FPN	mAP	mAR	param	GFLOPs	FPS
		0.229	0.364	11.2M	28.6	328
√		0.232	0.369	9.8M	26.7	329
	√	0.236	0.375	7.9M	25.0	331
√	√	**0.241**	**0.378**	**6.5M**	**22.8**	**332**

TABLE VII
COMPARISON OF WGP

Models	mAP	mAR	param	GFLOPs	FPS
efficientViT[49]	0.180	0.308	**8.4M**	**20.6**	104
CSwinTransFormer	0.211	0.353	28.5M	83.2	46
Repvit[50]	0.216	0.356	11.1M	29.6	215
WGP	**0.232**	**0.369**	9.8M	26.7	**329**

TABLE VIII
COMPARISON OF RFS-FPN

Models	mAP	mAR	param	GFLOPs	FPS
BIFPN	0.225	0.365	7.4M	25.0	322
AFPN	0.212	0.348	8.9M	25.1	190
EfficientRep[51]	0.222	0.360	10.2M	25.6	**331**
HS-FPN	0.210	0.349	**7.1M**	**23.9**	310
MAFPN[52]	0.228	0.368	11.2M	31.2	310
RFS-FPN	**0.236**	**0.375**	7.8M	25.0	**331**

conducted ablation experiments on the VisDrone dataset using YOLOv8s as the baseline, as shown in Table VI. By replacing the corresponding structures in YOLOv8s with WGP and RFS-FPN modules individually, the results demonstrate that each module significantly improves both mAP and mAR metrics while substantially reducing total parameters and GFLOPs. Overall, the mAP and mAR increased to 0.241 and 0.378, respectively, while the parameters and GFLOPs decreased to 6.5M and 22.8, respectively.

Further comparison experiments on other models were performed to evaluate the generalization capability of these modules, as shown in Tables VII and VIII. The results show that in lightweight models, WGP and RFS-FPN achieve notable detection accuracy improvements with minimal parameter increases, while, in high-performance models, they enhance precision while reducing parameters. Overall, both the integrated model EWS-YOLO and the standalone modules exhibit efficient and consistent performance gains, confirming strong compatibility and the absence of design conflicts between the modules.

2) Effect of AHC and DROP to WGP: To validate the effectiveness of the AHC and DROP modules compared to the WGP architecture, we conducted systematic ablation experiments on the YOLOv8s baseline model, as shown in Table IX. When the AHC module was enabled, the model achieved the highest mAP (0.233) and the lowest number of parameters (9.7M), demonstrating its key role in improving detection accuracy and model lightweighting. When only the DROP module was enabled, the model performed optimally in computational efficiency, with the lowest GFLOPs (24.7) and the highest FPS (372), indicating its effectiveness in reducing computational overhead. When both AHC and DROP were used together, the model still achieved a significant reduction in both the number of parameters (9.8M) and computational cost (26.7 GFLOPs), with the inference speed (329 FPS) remaining at a high level. This indicates that the AHC and DROP modules have complementary functions, and their synergistic use achieves an optimal balance among accuracy, lightweighting, and inference efficiency.

TABLE IX
EFFECT OF AHC AND DROP ON THE VISDRONE DATASET

AHC	DROP	mAP	mAR	param	GFLOPs	FPS
		0.229	0.364	11.2M	28.6	328
√		**0.233**	**0.372**	**9.7M**	29.8	304
	√	0.228	0.368	10.1M	**24.7**	**372**
√	√	0.232	0.369	9.8M	26.7	329

TABLE X
EFFECT OF PSU AND DFF ON THE VISDRONE DATASET

PSU	DFF	mAP	mAR	param	GFLOPs	FPS
		0.229	0.364	11.2M	28.6	328
√		0.234	0.372	11.4M	30.3	325
	√	0.231	0.368	**7.8M**	**24.7**	**342**
√	√	**0.236**	**0.375**	7.9M	25.0	331

3) Effect of PSU and DFF to RFS-FPN: To validate the effectiveness of the PSU and DFF modules compared to the RFS-FPN architecture, we conducted systematic ablation experiments on the YOLOv8s baseline model, as shown in Table X. The results demonstrate that replacing the original upsampling module with PSU increased the model's mAP by 0.5% points while adding only 0.2M parameters, significantly enhancing multiscale feature reconstruction. Meanwhile, substituting the C2F structure with the DFF module reduced parameters by 2.4M and GFLOPs by 3.9 while simultaneously improving detection accuracy. When jointly deploying PSU and DFF, the total parameter count remained at the same level as using DFF alone (with only a 0.1M increase), while mAP improved by 1.1% points over the baseline. These results not only confirm the independent contributions of both modules in feature upsampling and cross-layer fusion but also validate their compatibility and synergistic effects through coordinated parameter and accuracy optimization.

4) Comparative Visualization of WGP, PSU, and DFF: Fig. 9 presents visual comparisons of feature processing effects with other methods for the WGP, PSU, and DFF modules through (a)–(c). Red boxes mark the original feature distributions before processing, while brown boxes display postprocessing feature morphologies. In Fig. 9(a), the WGP module compared to C2F employs feature weaving technology to enhance spatial consistency between dense region features and the original image, significantly mitigating representation degradation in deep networks. In Fig. 9(b), the PSU module compared to upsample (bilinear interpolation) adopts a PSU strategy, achieving higher detail retention rates in output

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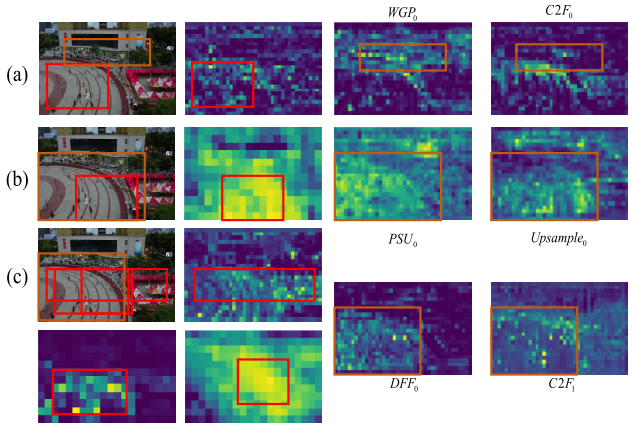


Fig. 9. Comparative visualization of WGP, PSU, and DFF. (a) Comparison between the WGP_0 and $C2F_0$ modules in processing feature maps. (b) Comparison between the PSU_0 and $Upsample_0$ modules in processing feature maps. (c) Comparison between the DFF_0 and $C2F_1$ modules in processing feature maps.

TABLE XI
SCORE COMPARISON OF THE VISDRONE DATASET

Models	Score
YOLOv8s	105.8
YOLOv10s	109.8
ASF-YOLO	88.1
Hyper-YOLO	81.4
GHOST-YOLO	-16.1
GOLD-YOLO	69.4
LSKNet	-37.4
Unireplknet	26.1
Starnet	24.5
EWS-YOLO	**122.9**
NanoDet	-98.6
PP-YOLO-TINY	-10.1
YOLOv8n	20.2
YOLOv10n	29.7
EWS-YOLOn	**128.6**

feature maps compared to traditional bilinear upsampling. In Fig. 9(c), the DFF module compared to C2F utilizes dynamic feature filtering to amplify target region feature responses while improving the signal-to-noise ratio of background interference, effectively eliminating redundancy in cross-layer fusion. Together, these modules collaboratively construct a multiscale optimization framework spanning local details to global semantics.

F. Significance Analysis

To demonstrate the significant performance improvement of EWS-YOLO, we conducted a significance analysis based on Table XI. In this analysis, the performance of each model is quantified by a composite score, calculated using the following formula:

$$N = \frac{(X - \text{Min}) \times 100}{\text{Max} - \text{Min}} \tag{24}$$

$$\text{PI} = N(\text{mAP}) + N(\text{mAP@.50}) + N(\text{mAP@.75}) + N(\text{FPS}) \tag{25}$$

$$\text{NI} = N(\text{Param}) + N(\text{GFLOPs}) \tag{26}$$

$$\text{Score} = \frac{\text{PI} - \text{NI}}{2} \tag{27}$$

TABLE XII
COMPARISON OF BRANCH RATIOS ON THE VISDRONE DATASET

$B_{(i,1)} : B_{(i,2)} : B_{(i,3)}$	mAP@.50	param
8:7:1	0.410	6.36
4:3:1	**0.424**	**6.52**
2:1:1	0.426	6.91
4:9:3	0.423	6.84
12:3:1	0.403	6.31

where N represents normalization, PI means positive indicators, and NI indicates negative indicators. This score definition effectively balances the model's accuracy and computational efficiency, providing a more comprehensive reflection of the model's overall performance. In the comparative experiments, both EWS-YOLO and EWS-YOLOn achieved the highest scores (all model scores are averages from ten measurements). To evaluate the statistical significance of their performance improvement, we conducted Shapiro–Wilk tests comparing EWS-YOLO with the second-best performing YOLOv10s and EWS-YOLOn with YOLOv10n, respectively. The null hypothesis was set as the performance improvement of EWS-YOLO and EWS-YOLOn over YOLOv10s and YOLOv10n is not significant. The test results showed that the p-values for both comparisons were less than 0.05, leading to a statistically significant rejection of the null hypothesis. This confirms that the proposed models achieve a significant performance improvement compared to all the baseline models.

G. Sensitivity Analyses

1) Branch Ratio Analysis to WGP: To determine the optimal channel ratio of branches in the WGP module, we conducted a comparative experiment on the VisDrone dataset with different branch ratios, and the results are shown in Table XII. Based on the conventions of related model designs, we first set the channel number of branch $B_{(i,1)}$ to half of the total channels, with the remaining channels allocated to branches $B_{(i,2)}$ and $B_{(i,3)}$, and then conducted experimental comparisons with different ratios such as 7:1, 3:1, and 1:1. The experimental results show that when the proportion of branch $B_{(i,3)}$ is too low, the model performance significantly decreases. Under the setting where $B_{(i,2)}$: $B_{(i,3)}$ is 1:1, mAP@0.50 reaches 0.426, the highest among all ratios, but the parameter count also increases to 6.91M, which is about 7% higher than the 3:1 ratio. It is worth noting that when the ratio is adjusted from 7:1 to 3:1, mAP@0.50 increases from 0.410 to 0.424, a gain of 2.9%, while the parameter count only increases from 6.36 to 6.52M, a rise of about 2.5%, achieving a better balance between performance and model efficiency. Therefore, the optimal ratio between $B_{(i,2)}$ and $B_{(i,3)}$ is determined to be 3:1.

Further considering $B_{(i,2)}$ and $B_{(i,3)}$ as a whole compared to $B_{(i,1)}$, experiments show that when the channel ratio between $B_{(i,1)}$ and $B_{(i,2)} + B_{(i,3)}$ is 1:1 (corresponding to the overall ratio 4:3:1), mAP@0.50 reaches 0.424, the best result among all combinations. This ratio achieves high detection accuracy while controlling the parameter count, indicating that with the

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TABLE XIII
COMPARISON OF PSU AND UPSAMPLE

Models	mAP@.s	mAP@.M	mAP@.L
PSU	**0.141**	**0.360**	**0.501**
Upsample	0.135	0.347	0.462

overall channel allocation of 4:3:1, the model achieves a good balance between multibranch feature fusion and computational efficiency.

2) Analysis of Impact of Head Count on AHC: To determine the optimal number of heads in the head shuffle mechanism, we conducted comparative experiments with configurations of 4, 8, and 16 heads, achieving mAP@0.50 scores of 0.415, 0.424, and 0.417, respectively. The results indicate that an insufficient number of heads limits the effectiveness of the shuffle operation's interaction, while an excessive number leads to high similarity among attention heads, resulting in feature redundancy and computational waste. Considering the performance, we ultimately set the number of heads to 8.

3) Analysis of Middle Upsampling Ratio on PSU: Due to the model's adoption of 2× upsampling, and to prevent pixel loss caused by divisibility issues, the intermediate sampling values could only be chosen as 1.6× or 1.25×. We compared two sequences: 1.6× upsampling followed by 1.25× upsampling and 1.25× upsampling followed by 1.6× upsampling. The corresponding mAP@0.50 values were 0.424 and 0.419, respectively. Analysis indicates that performing the larger upsampling ratio first can better preserve deep feature information, while cooperating with the shuffle operation in the PSU can more fully utilize its channel shuffling capability. Therefore, we ultimately selected the strategy of performing 1.6× upsampling first, followed by 1.25× upsampling.

4) Comparison Analysis of PSU and Upsample: Table XIII compares the detection performance of PSU and bilinear interpolation (Upsample) on targets of different scales. It can be observed that PSU provides significant accuracy improvements for small-, medium-, and large-sized targets, with a particularly notable enhancement in the detection accuracy for small targets. Although Table X indicates a slight increase in parameters for PSU compared to upsample, Fig. 9(b) demonstrates that the results after PSU upsampling align more closely with the target annotation information.

V. CONCLUSION

In this article, we propose EWS-YOLO, an efficient object detection network based on feature weaving and redundancy suppression. The network employs the WGP module to achieve implicit multiscale feature extraction and adaptive fusion, significantly enhancing multiscale detection capability. In addition, the RFS-FPN integrates low-level spatial details with high-level semantic features while effectively suppressing redundant information through adaptive channelwise recalibration. Experiments on the VisDrone and DIOR datasets demonstrate that EWS-YOLO achieves 42.4% and 95.2% mAP@0.5, respectively, while reducing parameters by 41% compared to YOLOv8s, striking a remarkable balance between accuracy and efficiency for resource-limited scenarios. Ablation studies further validate the effectiveness of the proposed modules. Future work will focus on further improving the detector's efficiency–performance tradeoff.

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